

Suggested coding solution Spring 2020

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#-----WD
setwd("C:/Users/s13163/Dropbox/FIE401 exam")

#-----Packages
library(readr)
library(lubridate)
library(dplyr)
library(DescTools)
library(psych)
library(xtable)
require(DescTools)
require(tibble)
require(knitr)
require(MatchIt)
require(lmtest)
require(sandwich)
require(car)
require(mfx)
require(stargazer)
require(AER)
require(plm)

#-----Data import and merge
event <- read_csv("Dates_Hybrid_Intro.csv")
#TICKER: firm identifier
#date: date in yyyymmdd format
#Eventtime
# for each firm, there are different event dates!
# EventTime=0 is the rollover date for a particular firm

activity <- read_csv("HFT_proxy.csv")
#TICKER: firm identifier
#DATE: date (in yyyymmdd format)
#QTE_DVOL: number of quote updates to the dollar volume executed per USD 10,000.

load("Data_from_CRSP.RData")
crsp=Data_from_CRSP
rm(Data_from_CRSP)

# split dates
event$date=as.character(event$date) # must be first changed into character
event$date=as.Date(event$date, format="%Y%m%d")
activity$DATE=as.character(activity$DATE) # must be first changed into character
activity$DATE=as.Date(activity$DATE, format="%Y%m%d")
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crsp$date=as.character(crsp$date) # must be first changed into character
crsp$date=as.Date(crsp$date, format="%Y%m%d")

m=merge(activity, event, by.x=c("TICKER", "DATE"), by.y=c("TICKER", "date"),
        all.x=T, all.y=F)

# now merge with CRSP
m=merge(m, crsp, by.x=c("TICKER", "DATE"), by.y=c("TICKER", "date"), all.x=T, all.y=F)

#-----Post dummy
m$post=ifelse(m$EventTime>=0,1,0)

#-----Clean up the data
m<-m[m$VOL>0,] #keep only days with non-zero trading volume
m$ASKHI[m$PRC<0]<-NA
m$BIDLO[m$PRC<0]<-NA
m$PRC<-abs(m$PRC)

#--Removing missing observations
m<-na.omit(m)
#--Exclude stocks that trade less than 200 days in our sample period
#--and that have price below 5 USD or above 1,000 (classical price filters)
nobs<-m %>%
  group_by(TICKER) %>%
  summarise(N=n(),AVPRC=mean(PRC)) %>%
  filter(N>=200 & AVPRC>=5 & AVPRC<=1000)

m<-merge(m,nobs,by="TICKER")

#-----Volatility measure
m$RANGE<-100*sqrt(((log(m$ASKHI)-log(m$BIDLO))^2)/(4*log(2)))
# high minus low range as in Parkinson (1980)
m$SDRET<-100*rollapply(m$RET,FUN=sd,na.rm=TRUE,width = 5,fill = NA, align = 'r')
# rolling standard deviation of returns

#-----Control variables
m$mcap=abs(m$PRC)*m$SHROUT*1000 # shROUT is in 1000.
m$mcap=log(m$mcap/1000000000) # log(mcap)
m$bidask=(-m$BID+m$ASK)*2/(m$BID+m$ASK) # proportional quoted spread
m$turn<-(m$VOL/m$SHROUT)*(252) # annualized turnover
m$inv_prc<-1/m$PRC # price inverse

m<-m[,c("TICKER","DATE","post","mcap","bidask","turn","RANGE","PRC",
        "inv_prc","QTE_DVOL","SDRET")]

#-----Winsorize
m$QTE_DVOL <- Winsorize(m$QTE_DVOL, probs = c(0.005, 0.995), na.rm=T)
m$RANGE <- Winsorize(m$RANGE, probs = c(0.005, 0.995), na.rm=T)
m$SDRET <- Winsorize(m$SDRET, probs = c(0.005, 0.995), na.rm=T)
m$bidask <- Winsorize(m$bidask, probs = c(0.005, 0.995), na.rm=T)

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m$mcap <- Winsorize(m$mcap, probs = c(0.005, 0.995), na.rm=T)
m$inv_prc <- Winsorize(m$inv_prc, probs = c(0.005, 0.995), na.rm=T)
m$turn <- Winsorize(m$turn, probs = c(0.005, 0.995), na.rm=T)

#-----Part 1 Summary statistics-----#

m2=subset(m, select=c("QTE_DVOL", "RANGE", "SDRET","bidask", "mcap","inv_prc","turn"))
sum=describe(m2)
sum2=subset(sum, select=c("n", "mean", "median", "sd", "min", "max"))
xtable(sum2)

#-----Part 2-----#

#-----Declare a panel
m<-pdata.frame(m,index=c("TICKER", "DATE"));
pdim(m);

#-----Regressions
#in case of staggered rollover, running a panel regression with twoway fixed effects
#is essentially implicit difference-in-difference approach
#when stock that are not yet rolled over to a new system are compared
#to stocks that are already rolled over
fit_1 <- plm(QTE_DVOL~post, data = m,
             model = "within",
             effect = "twoways")
se_fit_1 <- coeftest(fit_1, vcov = vcovHC(fit_1,cluster="group",type="sss"))[,2]

fit_2 <- plm(QTE_DVOL~post+bidask+mcap+inv_prc+turn, data = m,
             model = "within",
             effect = "twoways")
se_fit_2 <- coeftest(fit_2, vcov = vcovHC(fit_2,cluster="group",type="sss"))[,2]

#-----F-test
myHO <- c("post=0")
linearHypothesis(fit_1,myHO,
                 vcov=vcovHC(fit_1,cluster="group",type="sss"))
linearHypothesis(fit_2,myHO,
                 vcov=vcovHC(fit_2,cluster="group",type="sss"))

#-----Stargazer
stargazer(list(fit_1, fit_2),
           coef = list(fit_1$coefficients, fit_2$coefficients),
           se = list( se_fit_1, se_fit_2),
           type="text",
           keep.stat=c("adj.rsq"),
           report=('vc*t'),
           add.lines = list(
             c("Observations",
               length(fit_1$residuals), length(fit_2$residuals)),
             c("Model", "OLS", "OLS"),
             c("Stock FE", "yes","yes"),
             c("Day FE", "yes","yes"),
             c("Cluster SE", "by stock","by stock"),

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        c("F-stat: post=0", "142.36", "197.73")
    ))

#-----Part 3-----#

#-----Regressions
#---OLS
fit_ols_1<-plm(SDRET~QTE_DVOL+bidask+mcap+inv_prc+turn,
              model="within",effect="twoway",data=m)
se_fit_ols_1 <- coeftest(fit_ols_1,
                       vcov = vcovHC(fit_ols_1,cluster="group",type="sss"))[,2]

fit_ols_2<-plm(RANGE~QTE_DVOL+bidask+mcap+inv_prc+turn,
              model="within",effect="twoway",data=m)
se_fit_ols_2 <- coeftest(fit_ols_2,
                       vcov = vcovHC(fit_ols_2,cluster="group",type="sss"))[,2]

#---IV
fit_iv_1<-plm(SDRET~QTE_DVOL+bidask+mcap+inv_prc+turn | post+bidask+mcap+inv_prc+turn,
              model="within",effect="twoway",data=m)
se_fit_iv_1 <- coeftest(fit_iv_1 ,
                       vcov = vcovHC(fit_iv_1,cluster="group",type="sss"))[,2]

fit_iv_2<-plm(RANGE~QTE_DVOL+bidask+mcap+inv_prc+turn | post+bidask+mcap+inv_prc+turn,
              model="within",effect="twoway",data=m)
se_fit_iv_2 <- coeftest(fit_iv_2 ,
                       vcov = vcovHC(fit_iv_2,cluster="group",type="sss"))[,2]

#-----Stargazer
stargazer(list(fit_ols_1, fit_ols_2,fit_iv_1, fit_iv_2),
          coef = list(fit_ols_1$coefficients, fit_ols_2$coefficients,
                      fit_iv_1$coefficients, fit_iv_2$coefficients),
          se = list(se_fit_ols_1, se_fit_ols_2,
                   se_fit_iv_1, se_fit_iv_2 ),
          type="text",
          keep.stat=c("adj.rsq"),
          report=('vc*t'),
          add.lines = list(
            c("Observations",
              length(fit_ols_1$residuals), length(fit_ols_2$residuals),
              length(fit_iv_1$residuals), length(fit_iv_2$residuals)),
            c("Model", "OLS", "OLS", "2SLS", "2SLS"),
            c("Stock FE", "yes", "yes", "yes", "yes"),
            c("Day FE", "yes", "yes", "yes", "yes"),
            c("Cluster SE", "by stock", "by stock", "by stock", "by stock")
          )
    ))

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